

Hierarchical Balance Packing: Towards Efficient Supervised Fine-tuning for Long-Context LLM

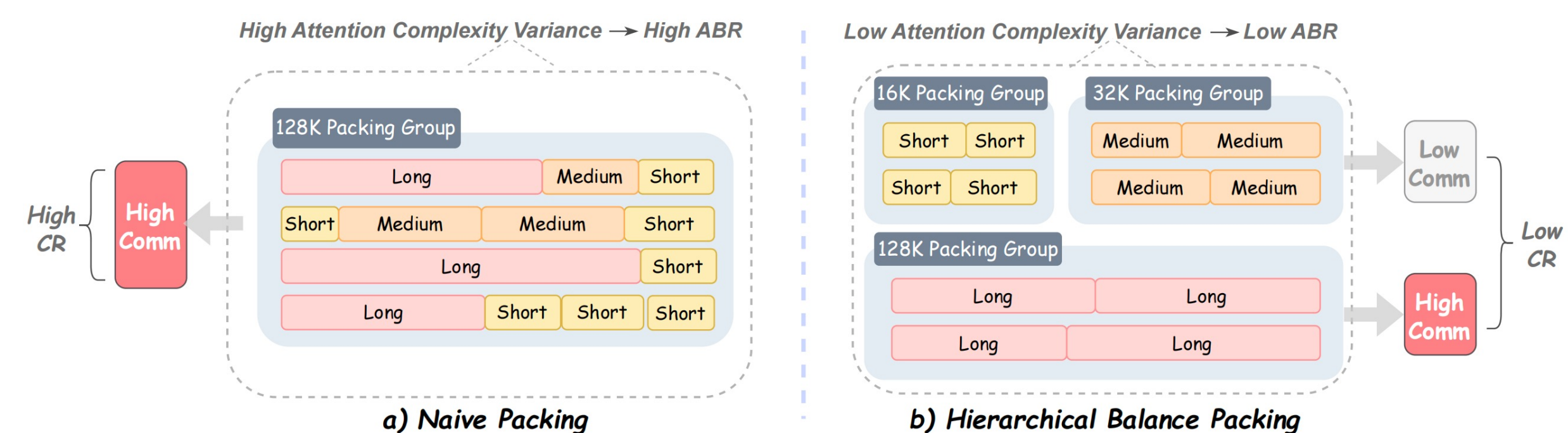
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Contributions

- **Hierarchical Balance Packing (HBP):** Multi-level packing to balance attention and reduce communication.
- **Dynamic Training Pipeline:** Adaptive parallelism + curriculum learning + stable loss.
- **Efficiency & Performance:** $1.4\times\text{--}2.4\times$ faster training while preserving accuracy.
- **Practical:** Overhead $<2\%$ of total training time.

Problem & Motivation



➤ Workload imbalance

- Excessive padding in mini-batches $\rightarrow$ wasted computation.
- Uneven workload across devices $\rightarrow$ idle GPU time.

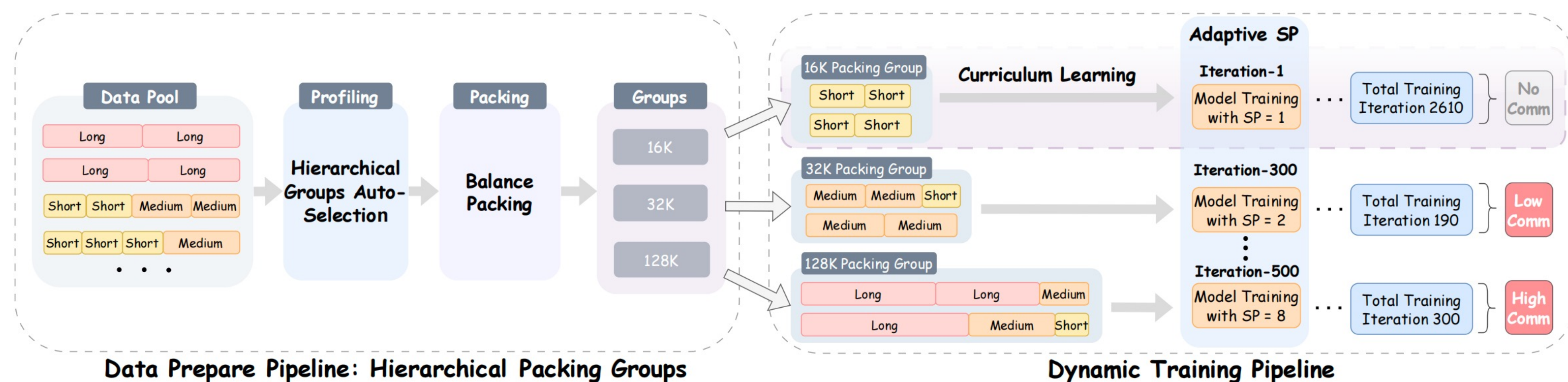
➤ Attention imbalance

- Naive packing mixes short and long sequences $\rightarrow$ high variance in attention complexity.

➤ Communication overhead

- Long sequences require sequence parallelism (SP).
- Mixing with short sequences forces **extra communication**.

Framework



Auto-selection

- Profile candidate lengths (8K-128K).
- Find best SP + **GC config** per length.
- Optimize groups to reduce communication.

Balance Packing

- Assign samples to optimal group.
- **Greedy fill** reduces padding waste.
- Construct balanced batches by attention complexity.

Dynamic Pipeline

- **Adaptive SP:** switch configs with zero overhead.
- **Curriculum Learning:** short first $\rightarrow$ hybrid later.
- **Stable Loss:** Ensure equal token contributions.

Experiments

Model (Type)	General Tasks							Long Tasks				GPU Days
	AVE	MMLU	BBH	IFEval	Math	GSM8k	HumanEval	Ruler (32K 128K)	LongBench	LongCite	(speed up)	
LLama3.1-8B												
LongAlign-packing	56.6	44.5	65.5	67.8	30.7	80.6	62.2	84.5 57.5	46.7	67.8	5.4 (0.97×)	
LongAlign-sorted	57.6	62.7	65.4	67.8	32.8	82.2	61.6	85.8 59.9	46.5	64.0	33.3 (0.16×)	
ISF	56.0	54.5	65.3	70.4	33.7	81.7	62.8	85.0 67.4	44.0	71.6	5.22 (1.0×)	
HBP	58.2	63.0	67.2	67.7	33.0	81.9	63.4	85.6 70.8	43.1	71.5	3.73 (1.4×)	
Qwen2.5-32B												
ISF	73.5	74.8	83.5	75.6	56.5	93.7	86.6	88.2 59.3	51.0	60.2	21.3 (1.0×)	
HBP	76.2	76.6	83.6	76.0	57.1	94.4	84.2	88.3 59.0	51.9	61.7	16.0 (1.33×)	
LLama3.1-70B												
ISF	72.1	78.9	83.0	77.6	44.1	85.3	76.8	91.8 57.1	50.4	72.7	44.4 (1.0×)	
HBP	74.2	81.5	83.1	76.2	48.3	93.3	77.4	93.4 57.5	52.2	75.3	31.1 (1.42×)	
DeepSeek-V2 (236B)												
ISF	71.8	76.8	84.0	71.1	41.3	89.0	78.6	86.6 -	47.1	-	57.1 (1.0x)	
HBP	72.0	76.5	83.1	72.6	41.4	89.9	78.1	87.3 -	50.3	-	23.8 (2.4×)	

Model	CL	AVE	LongBench
LLama3.1-8B-HBP		56.6	41.6
LLama3.1-8B-HBP	✓	58.2	43.1
LLama3.1-70B-HBP		72.2	51.5
LLama3.1-70B-HBP	✓	74.2	52.2
LLama3.1-8B-ISF		56.0	44.0
LLama3.1-8B-ISF	✓	57.4	43.4

Model	CL-Iterations	AVE	LongBench
LLama3.1-8B-HBP	0	56.6	41.6
LLama3.1-8B-HBP	100	58.0	43.1
LLama3.1-8B-HBP	500	58.2	43.1
LLama3.1-70B-HBP	0	72.2	51.5
LLama3.1-70B-HBP	100	73.8	52.4
LLama3.1-70B-HBP	500	74.2	52.2

Loss Normalizer	AVE	LongBench	Ruler-128K	Longcite
Sum	56.7	42.5	65.2	70.5
Sample-Mean	55.5	42.9	46.1	70.3
Token-Mean	56.6	41.6	67.5	70.6
Ave-Token	57.6	43.1	70.8	71.2

HBP achieves up to $2.4\times$ faster training with no accuracy loss, stabilizes long-context SFT/RFT.